A Synthetic Ground-Truth Dataset for Online Map Matching

Exact segment-level ground truth for reproducible map-matching evaluation

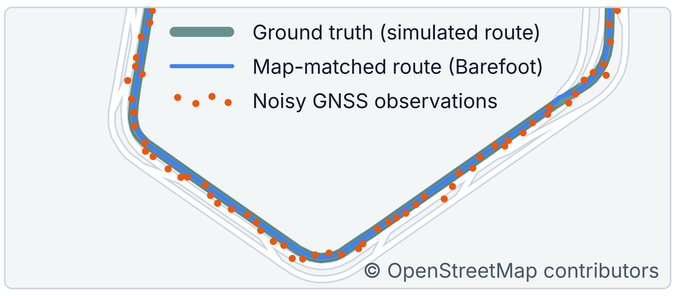
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1 The problem

Evaluating map matching requires ground truth, which real traces rarely provide [1].

Urban GNSS is noisy (multipath, NLOS) and often sparse. Map matching associates those points with road segments [3]. But real traces rarely say which ones were actually driven.



2 Our solution

KEY IDEA

Simulate the trip → keep the exact route → degrade only the observations.

A parameterizable, auditable pipeline from OpenStreetMap to segment-level ground truth. The O'Hare dataset is its demonstration instance.

A controlled benchmark for online map matching under spatial noise and temporal sparsity.

3 Pipeline

STEP 1

Network

`OSMnx [4] → osmfilter → netconvert (O'Hare, Chicago)`

→

STEP 2

Simulation

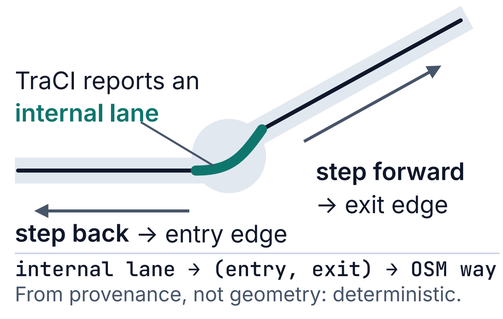
`randomTrips + SUMO [5] TraCI logs every 0.5 s seed 42, trips ≥ 500 m`

→

STEP 3

Reconciliation

`SUMO edge → OSM way deterministic via provenance metadata`



`internal lane → (entry, exit) → OSM way`

From provenance, not geometry: deterministic.

→

STEP 4

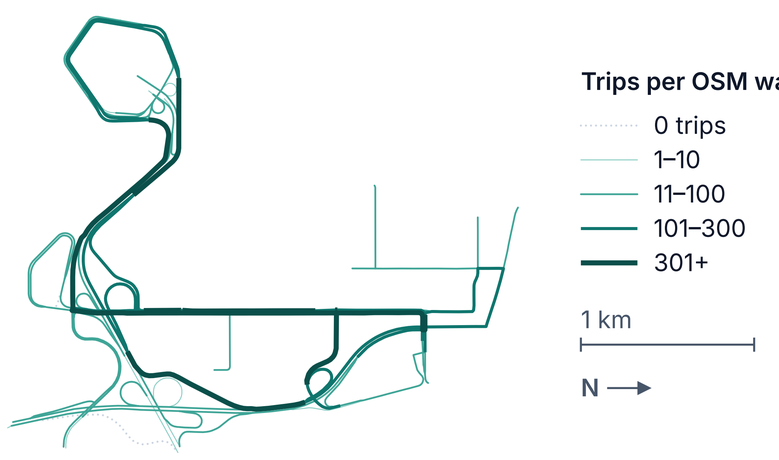
Degradation

`Gaussian noise σ = 5 m [1]`



Degrade observations, never the route. Every noisy and sparse variant inherits the same reference path, so quality changes come from noise or sparsity, not from the route.

4 Dataset at a glance



1,759 TRAJECTORIES

0.52–8.97 km ROUTE LENGTH

4 SAMPLING INTERVALS (1–20 s)

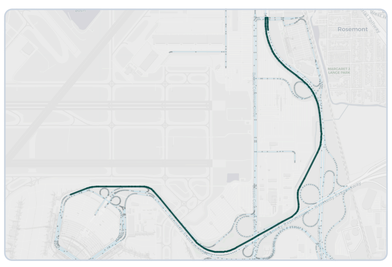
σ = 5 m GAUSSIAN GNSS NOISE

~440 trajectories per length quartile · stored as Parquet with OSM edge ids

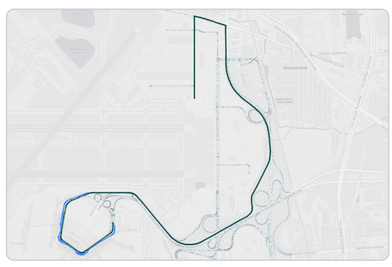
5a Sanity check

10 random routes (GraphHopper): 3 perfect · 3 minor · 4 significant.

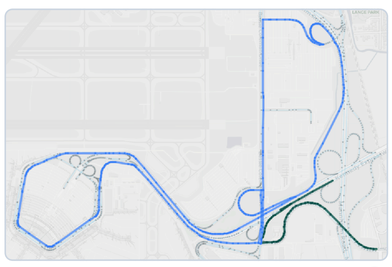
3 PERFECT



3 MINOR



4 SIGNIFICANT



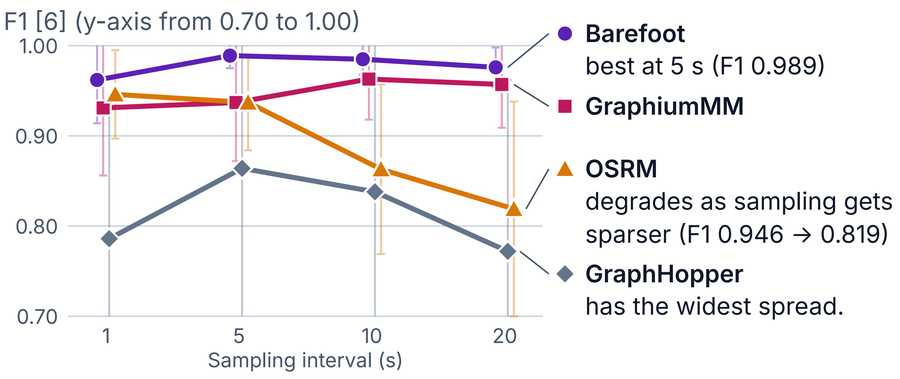
Basemap © OpenStreetMap contributors

Same legend as ①

The challenging cases (40%) occur near junctions, ramps and parallel roads, making the dataset a demanding benchmark.

5b Benchmark

Under identical routes and noise, the four tools perform distinctly.



20 routes (5 per quartile) · 790 / 800 valid executions · NKError [2] results in the paper.

6 Conclusion

- Exact segment-level ground truth, reproducible from seed to Parquet.
- Topology, spatial noise and temporal sparsity controlled independently.
- Structurally consistent and able to discriminate matchers.

Next: more cities, richer sensor-error models, latency benchmarks, real-trace comparison.

REFERENCES

[1] Erdmann & Ebendt (2014). Benchmarking Trajectory Matchers with SUMO. ESM.
[2] Newson & Krumm (2009). Hidden Markov map matching through noise and sparseness. SIGSPATIAL.
[3] Chao et al. (2020). A Survey on Map-Matching Algorithms. arXiv:1910.13065.
[4] Boeing (2025). Modeling and Analyzing Urban Networks and Amenities With OSMnx. Geogr. Anal.
[5] Alvarez Lopez et al. (2018). Microscopic Traffic Simulation using SUMO. IEEE ITSC.
[6] Li et al. (2014). Spatio-temporal trajectory simplification for inferring travel paths. SIGSPATIAL.

CODE + DATASET

`github.com/JoseEdSouza/map-matching`

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